編號： 198　　　　國立成功大學 113 學年度碩士班招生考試試題

系　所：電機資訊學院-資訊聯招

考試科目：計算機數學　　　　　　　　　　　　考試日期：0201，節次：3

第1頁，共2頁

※ 考生請注意：本試題不可使用計算機。 請於答案卷(卡)作答，於本試題紙上作答者，不予計分。

一、離散數學 (50%)

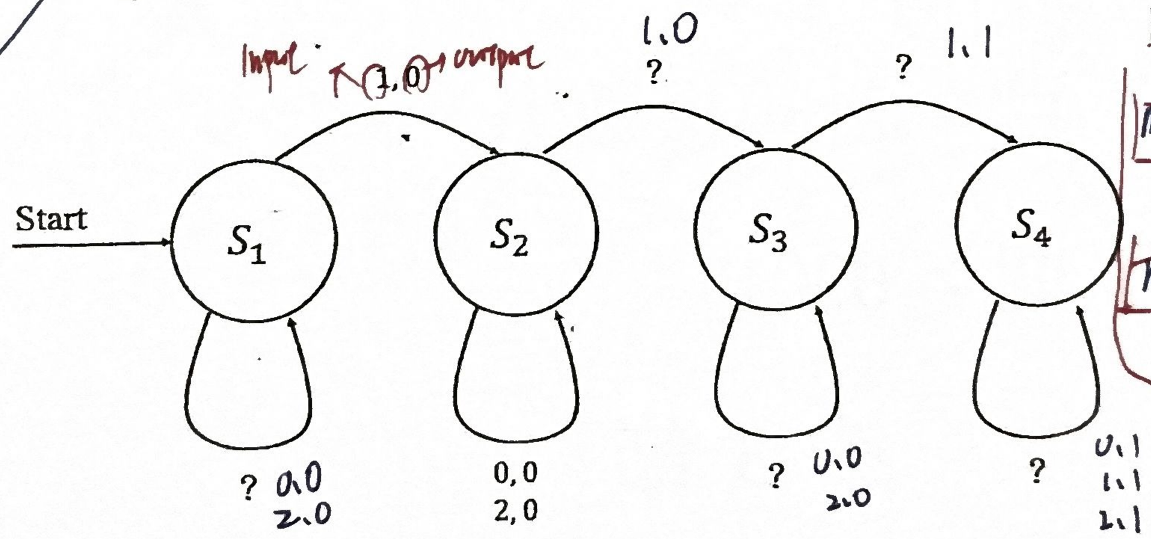
1. (7 points) Solve the recurrence relation $a_{n+2} - 4a_{n+1} + 3a_n = -200$ for $n \geq 0$ and $a_0 = 3000$ and $a_1 = 3300$.

2. (10 points) Please list the first 5 coefficients of the generating function $F(x) = \sqrt{1+x}$. (請化簡為最簡分數形式)

3. (15 points) If $G(x)$ is the generating function for the sequence $\{a_k\}$, what is the generating function for each of these sequences?
   (A) (5 points) $3a_0, 3a_1, 3a_2, 3a_3, \ldots$
   (B) (5 points) $0, 0, 0, 0, a_2, a_3, \ldots$
   (C) (5 points) $a_1, 2a_2, 3a_3, 4a_4, \ldots$

內部狀態 輸入 輸出

4. (13 points) Design a finite state machine $M = (S, \varphi, \sigma, v, \omega)$, where $S = \{s_1, s_2, s_3, s_4\}$, $\varphi = \{0,1,2\}$, $\sigma = \{0, 1\}$. The machine outputs 1 if the input string contains at least three 1s, otherwise it outputs 0. (請填問號應有的內容)



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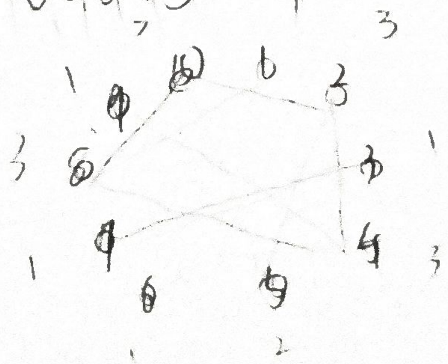
mealy ⇒ S×I ⇒ O
$v(s,2) = s_3$
moore ⇒ S → O
$w(s) = s_2$

5. (5 points) Consider the trees with vertices {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} that have corresponding degrees (1, 3, 1, 3, 2, 1, 1, 3, 1, 3). How many different spanning trees are there in total?

用 prufer 序列



⑤ 假設圖合法

利用 prufer 序列，有 n 頂點，prufer 序列長度為 $n-2$

頂點數 $n=10$

degree 序列：(1 3 1 3 2 1 1 3 1 3)

每個 vertex 在序列中出現的次數是 degree 減一

點	1	2	3	4	5	6	7	8	9	10
出現次數	0	2	0	2	1	0	0	2	0	2

$\boxed{\text{生成樹數量} = \text{prufer 序列的排列數}}$

故為 $\dfrac{8!}{2!\times 2!\times 1!\times 2!\times 2!} = 2520$ #

共 2520 種 #

完全圖 K_n 有 n^{n-2} 種 spanning tree

$K_{m,n}$ 有 $m^{n-1}\times n^{m-1}$ 種 spanning tree

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②

$F(x)=\sqrt{1+x}=(1+x)^{\frac{1}{2}}$，用二項式定理展開

其中 $\binom{\alpha}{k}$ 是廣義的定義為 $\dfrac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-k+1)}{k!}$

計算前5個係數 $\binom{\frac12}{0}$ $\binom{\frac12}{1}$ $\binom{\frac12}{2}$ $\binom{\frac12}{3}$ $\binom{\frac12}{4}$

$\binom{\frac12}{0}=1$，$\binom{\frac12}{1}=\dfrac{\frac12}{1!}=\dfrac12$，$\binom{\frac12}{2}=\dfrac{\frac12(\frac12-1)}{2!}=-\dfrac18$

$\binom{\frac12}{3}=\dfrac{\frac12(\frac12-1)(\frac12-2)}{3!}=\dfrac{1}{16}$，$\binom{\frac12}{4}=-\dfrac{5}{128}$

⑥ $\dim\left(\text{span}\left(\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix},\begin{bmatrix}x_2\\x_1\\x_3\\x_4\end{bmatrix},\ldots\right)\right)=3$

加入一個額外的線性關係，讓所有 vector 受限於該關係，進而降低維度

設 $x_1+x_2+x_3+x_4=0$ → 自由變數為3個

任何向量的排列也滿足這個條件

選 $x=(1\ 0\ 0\ -1)$ #

加要 $\dim=2$

⇒ 加 2 限制條件，使自由度變 2

$x_1+x_2=0$ $x_3+x_4=0$

取 $(1\ -1\ \ 1\ -1)$ #

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① $\lambda^2-4\lambda+3=0$, $\lambda=3$ or 1, $a_n^h=C_0 1^n+C_1 3^n$

$a_n^p=d_0 n$, 代回原式

$d_0(n+2)-4d_0(n+1)+3d_0 n=-2w$, $d_0=1w$

$a_n=C_0\times 1^n+C_1\times 3^n+1wn$, $C_0+C_1=30w$ $\quad C_1=1w$

$C_0+3C_1=320w$ $\quad C_0=29w$

$a_n=29w\times 1^n+1w\times 3^n+1wn$ #

③ A) $3G(x)$

B) $x^2G(x)-a_0x^2-a_1x^3$

D) $\frac{d}{dx}(G(x)-a_0)$

⑤ 沒有 spanning tree，因奇數 degree 的點必須偶數個，不是圖

⑥ $x=(1100)$

⑦ $G_4=\begin{vmatrix}0&1&1&1\\1&0&1&1\\1&1&0&1\\1&1&1&0\end{vmatrix}=\begin{vmatrix}3&1&1&1\\0&-1&0&0\\0&0&-1&0\\0&0&0&-1\end{vmatrix}=3\times(-1)^3$，可知用同樣方法

$G_n=(n-1)\times(-1)^{n-1}$

Chrys culture

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不予計分。

$\exists 0$

$)=S_3$

0

$=S_2$

⑥ $\dim(\ldots, [x_1], [x_2]) \longrightarrow$

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⑧ (B) $AB(B^{-1}) = BA$

AB 與 BA 相似

$AB = PDP^{-1}$ $\quad BA = BABB^{-1} = BPDP^{-1}B^{-1}$

$= (BP)D(BP)^{-1}$

故 eigen value of $AB = BA$ #

⑨ $\vec{PQ} = (3, 1, -2)$ $\vec{PS} = (a-5, b-1, c-1)$

$a-5 = 3r$ $\quad a = -1$, $r = -2$

$b-1 = -2$, $b = -1$

$c-1 = 4$, $c = 5$ $\quad (-1, -1, 5)$#

⑩ T，因是正相矩陣⇒能單式對角化
，所有λ均≥0，det≠0 故可逆

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二、線性代數 (50%)

6. (10%) Let a vector $\boldsymbol{x} = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$. It has 24 rearrangements like (x_1, x_2, x_3, x_4) and (x_4, x_3, x_1, x_2). Those 24 vectors span a subspace S. Find specific vectors $\boldsymbol{x}$ so that the dimension of S is three.

7. (10%) Let G_2 G_3 and G_4 be the determinants of the matrices in the following form:

$$G_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}, \quad G_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix}, \quad G_4 = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{vmatrix}$$

3 1 1 1   3 1 1 1
3 0 1 1   0 -1 0 0
3 1 0 1   0 0 -1 0
3 1 1 0   0 0 0 -1

-1   2   3

Calculate the value of G_n.

8. (10%) Let A and B as two matrices. If B is invertible, prove that AB has the same eigenvalues as BA.

9. (10%) Consider the points $P(3,-1,4)$ and $Q(6,0,2)$, and $R(5,1,1)$. Find the point S in $\mathbb{R}^3$ whose first component is -1 and such that $\overrightarrow{PQ}$ is parallel to $\overrightarrow{RS}$.

S(a, b, c)

(3 -1 -2)   (a-5, b-1, c-1)

10. True or False

(a) (2%) Every positive definite matrix is invertible.

(b) (2%) The determinant of $A - B$ equals $\det A - \det B$.

(c) (2%) If $\boldsymbol{u}$ is orthogonal to every vector of a subspace W, then $\boldsymbol{u} = \boldsymbol{0}$.

(d) (2%) If A is square and $A\boldsymbol{x} = \boldsymbol{b}$ is inconsistent for some vector $\boldsymbol{b}$, then the nullity of A is zero.

(e) (2%) If there is a basis for $\mathbb{R}^n$ consisting of eigenvectors of an $n \times n$ matrix A, then A is diagonalizable.

det(A-B)
det(A)-det(B)
Ax=0

↳要對角化要有n個獨立的eigen vector